

SmolVLM for Dense Video Captioning

Final Internship Update

Models Comparison

- Uniform Sampling - maximum number of frames: Qwen2.5=600, SmolVLM, InternVL3 and Ovis2=96
- MaxInfo: R=8, tol=0.3, initial number of frames=1000, maximum number of frames=the same as for FPS
- CSTA: initial number of frames=1000, maximum number of frames=the same as for FPS

Video Type	First	Center	FPS	MaxInfo	CSTA
Qwen2.5	41.9%	43.5%	60.4%	57.5%	59.4%
SmolVLM	36.2%	38.4%	50.7%	48.7%	49.2%
InternVL3	37.9%	42.8%	54.6%	53.6%	53.3%
Ovis2	40.8%	42.1%	54.2%	52.5%	53.3%

Table 1: VideoMME results over frame sampling techniques

Qwen2.5-VL-3B-Instruct

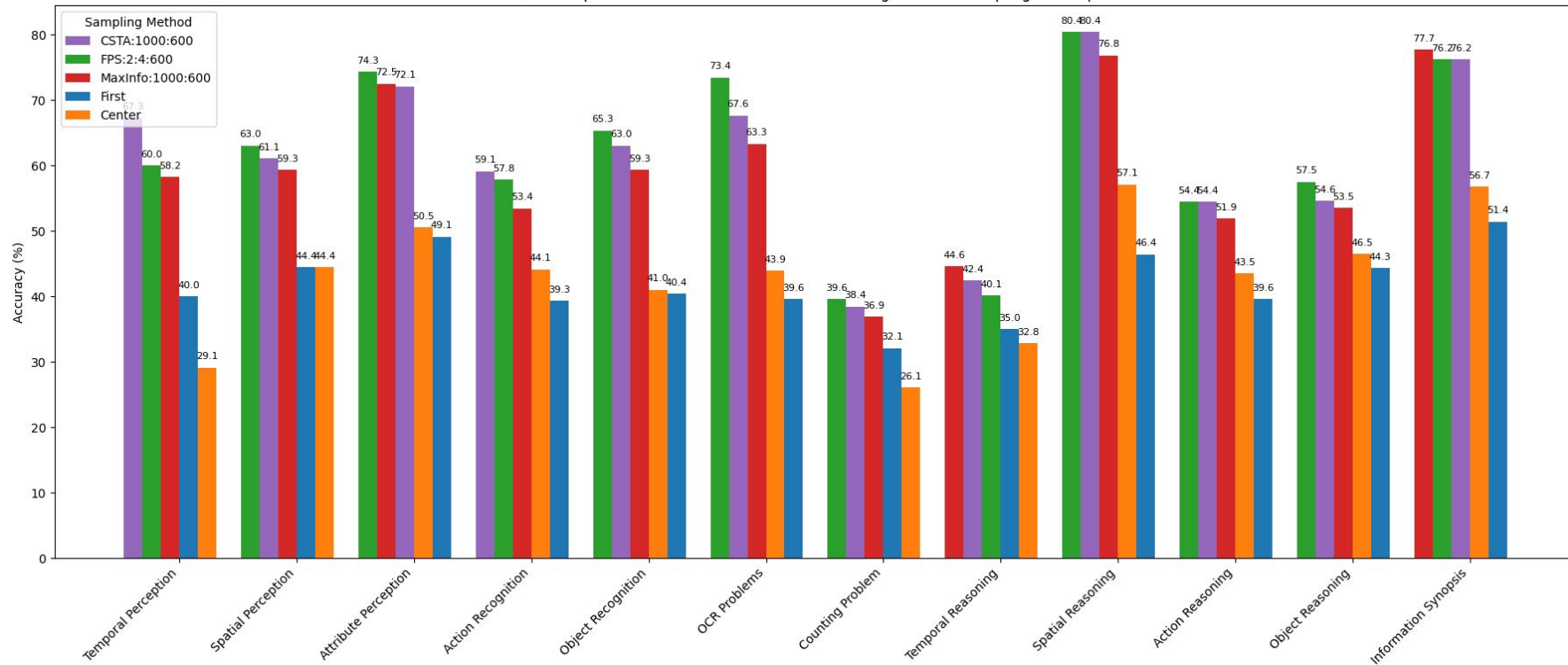
- MaxInfo:1000:600, average number of frames:87.67, execution time:75062.48s
- CSTA:1000:600, average number of frames:148.92, execution time:80961.79s
- FPS:2:4:600, execution time:82485.69s
- VideoMME result from the paper = 61.6% with 768 frames (2 FPS)

Video Type	First	Center	FPS:2:4:600	MaxInfo:1000:600	CSTA:1000:600
Short	46.0%	47.9%	72.6%	68.1%	70.2%
Medium	40.1%	40.0%	58.7%	54.9%	57.7%
Long	39.4%	42.6%	49.9%	49.6%	50.3%
Entire dataset	41.9%	43.5%	60.4%	57.5%	59.4%

Table 1: VideoMME results over frame sampling techniques: Model: Qwen2.5

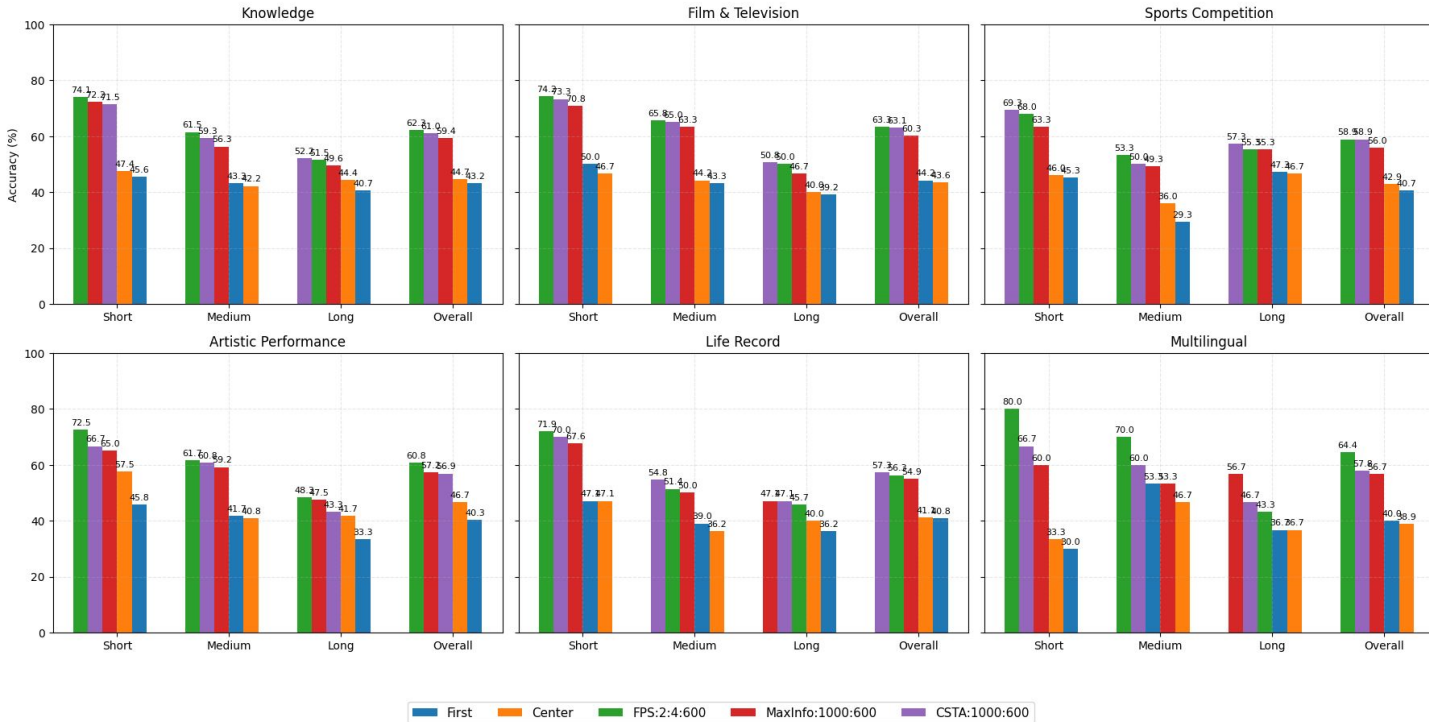
Qwen2.5-VL-3B-Instruct

Qwen2.5 performance on VideoMME for Taks Categories and Sampling Techniques



Qwen2.5-VL-3B-Instruct

Qwen2.5 performance on VideoMME for Video Categories and Sampling Techniques



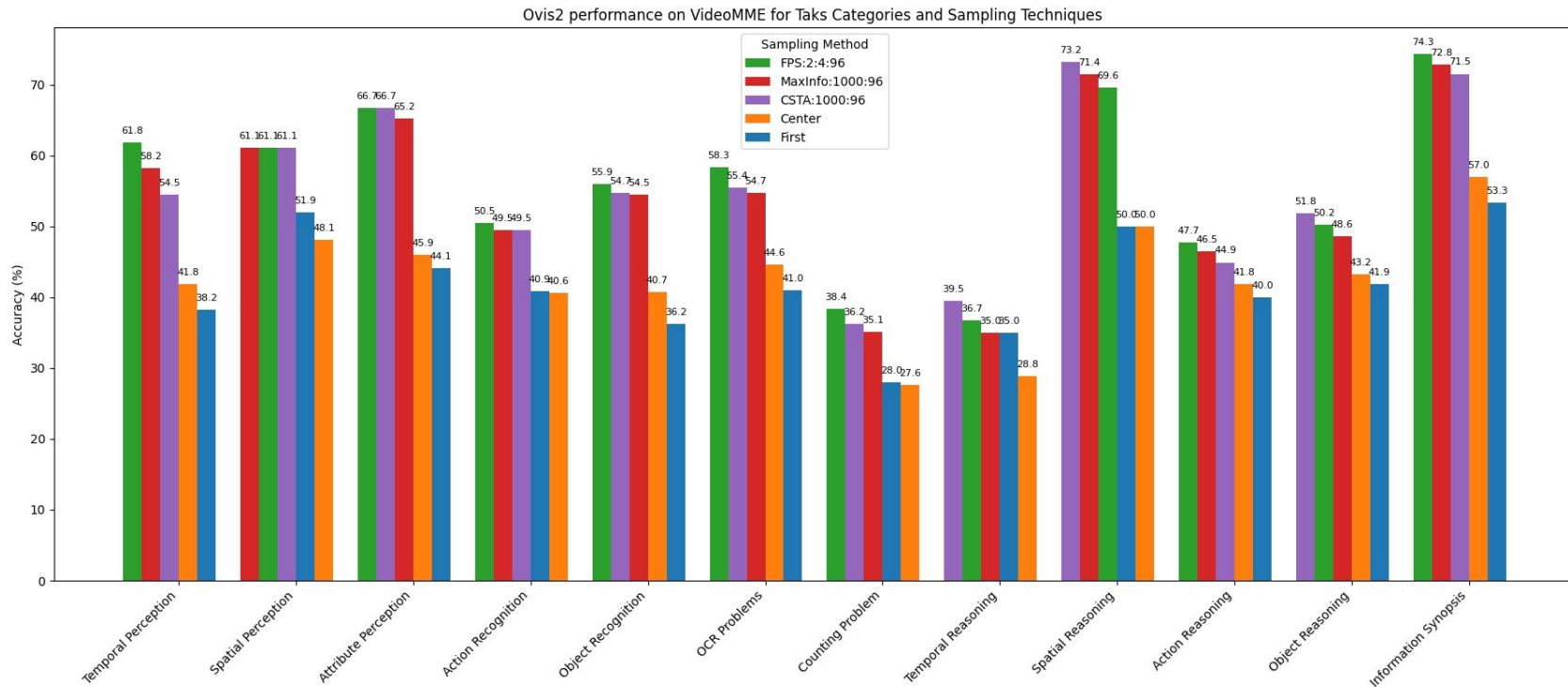
AIDC-AI/Ovis2-2B

- MaxInfo:1000:96, average number of frames:87.67
- CSTA:1000:96, average number of frames:95.76
- FPS:2:4:96
- VideoMME result from the paper = 57.2%, uniform sampling

Video Type	First	Center	FPS:2:4:96	MaxInfo:1000:96	CSTA:1000:96
Short	44.9%	45.8%	64.2%	62.6%	63.0%
Medium	39.4%	42.0%	53.0%	51.1%	51.7%
Long	38.0%	38.4%	45.3%	43.9%	45.2%
Entire dataset	40.8%	42.1%	54.2%	52.5%	53.3%

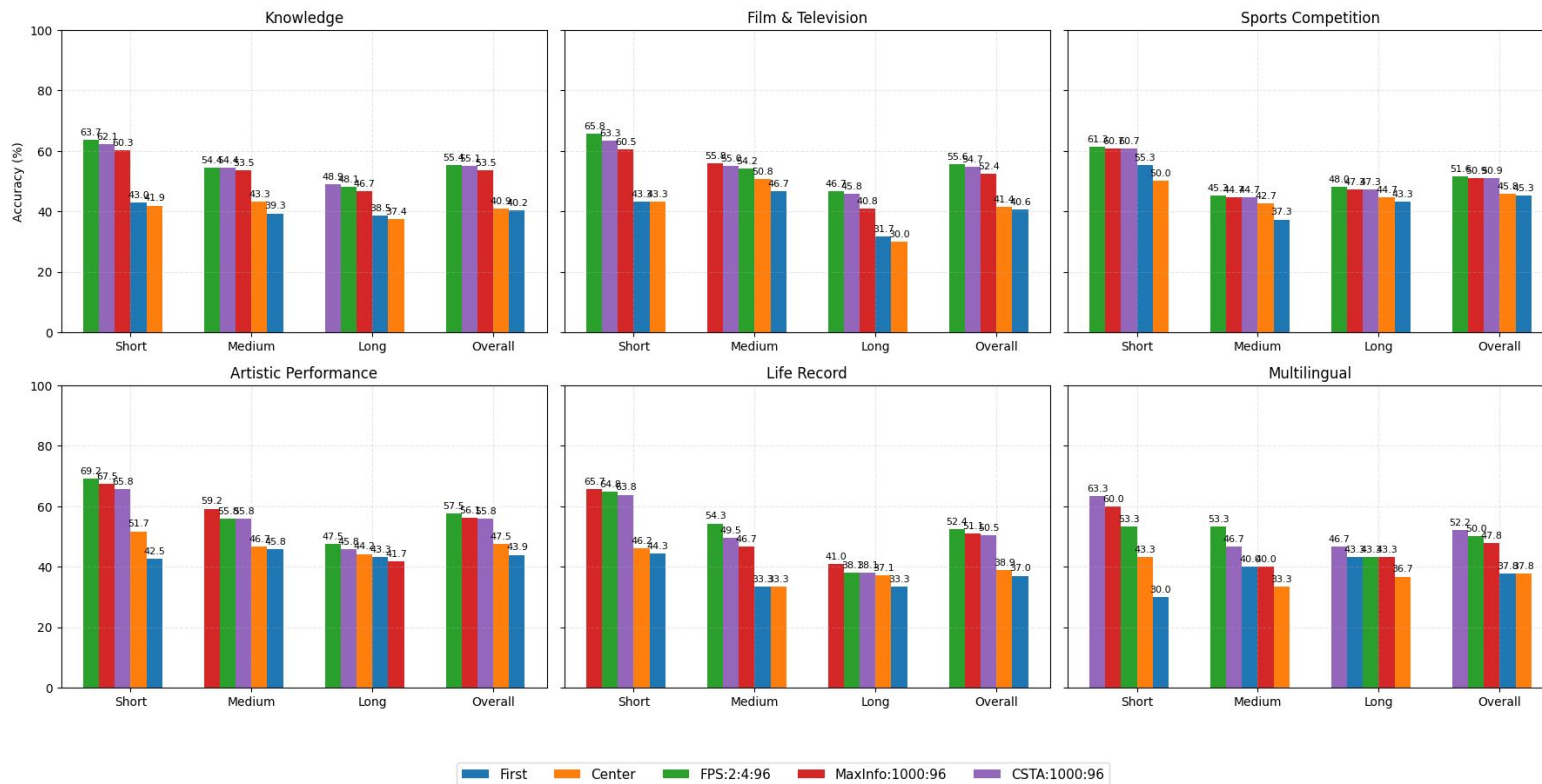
Table 3: VideoMME results over frame sampling techniques: Model: Ovis2

AIDC-AI/Ovis2-2B



AIDC-AI/Ovis2-2B

Ovis2 performance on VideoMME for Video Categories and Sampling Techniques



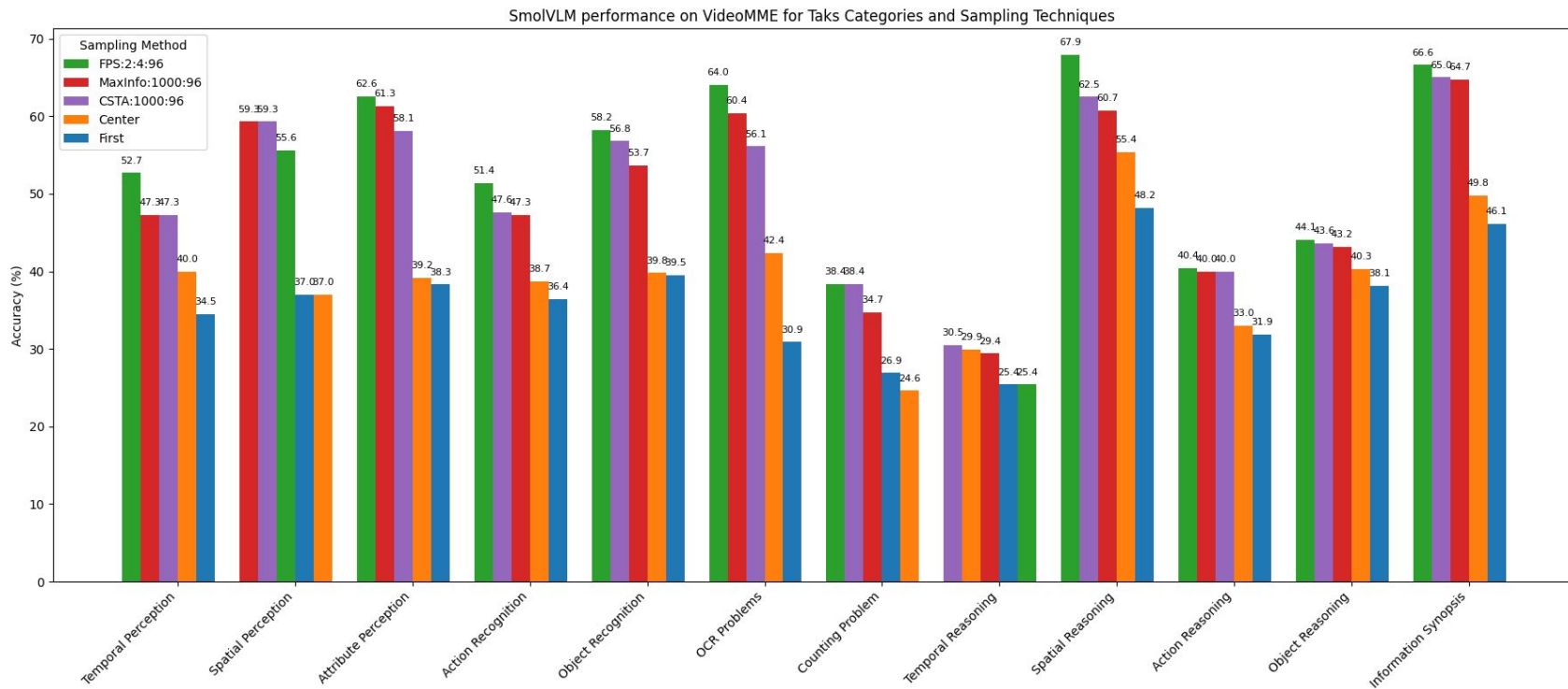
SmolVLM2-2.2B-Instruct

- VideoMME result from the paper = 52.1%, uniform sampling (1 FPS, maximum 50 frames)

Video Type	First	Center	FPS:2:4:120	FPS:1:4:120	FPS:2:4:96	FPS:1:4:50	MaxInfo:1000:120	CSTA:1000:120	CSTA:1000:96
Short	38.4%	41.4%	50.1%	60.4%	62.2%	63.6%	58.6%	48.6%	60.1
Medium	37.4%	40.4%	37.2%	37.2%	50.2%	51.3%	48.6%	35.6%	47.8
Long	32.8%	33.4%	32.4%	32.4%	39.8%	38.9%	38.9%	31.3%	39.8
Entire dataset	36.2%	38.4%	39.9%	43.4%	50.7%	51.3%	48.7%	38.5%	49.2

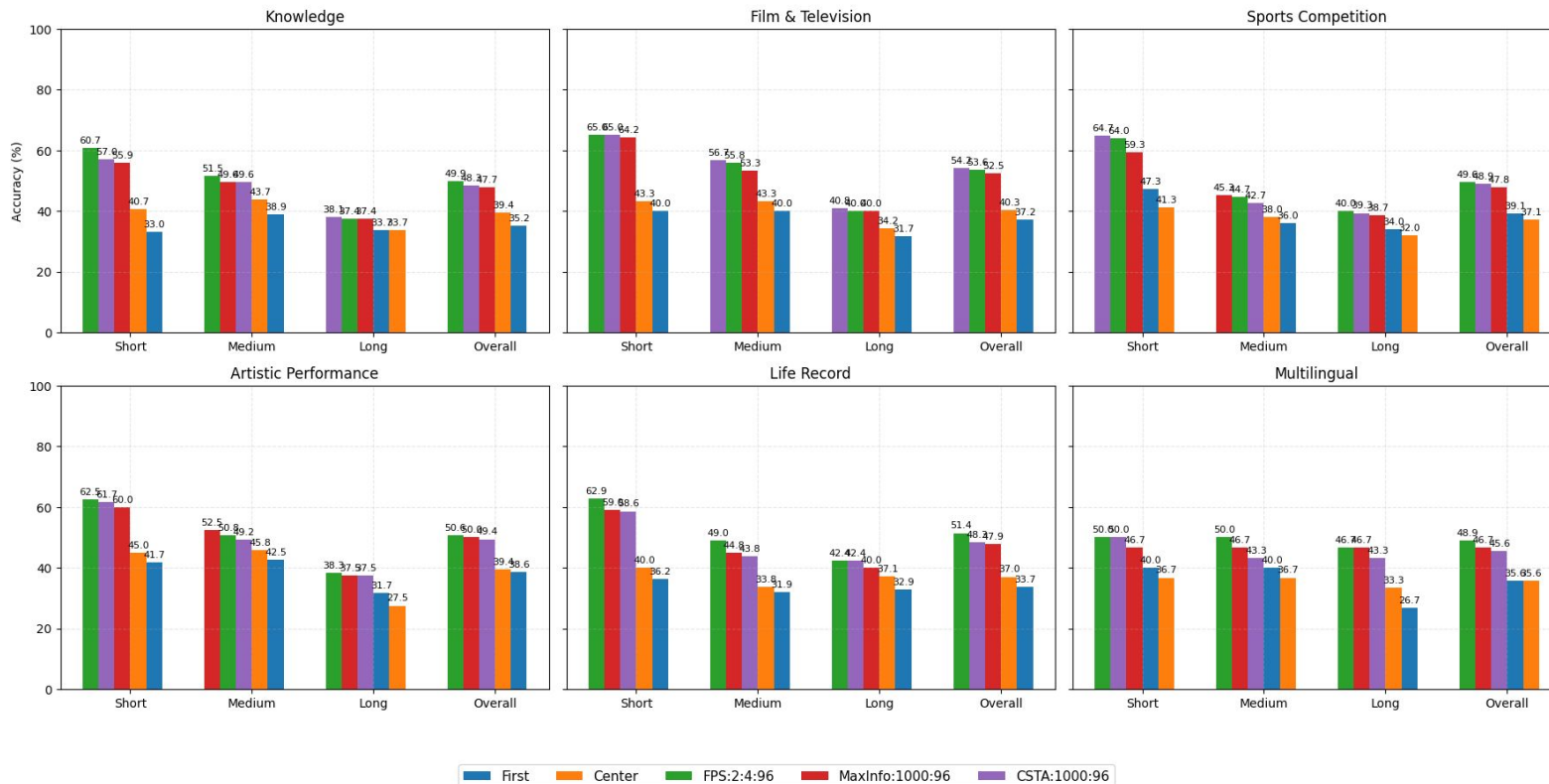
Table 5: VideoMME results over frame sampling techniques: Model: SmolVLM

SmolVLM2-2.2B-Instruct



SmolVLM2-2.2B-Instruct

SmolVLM performance on VideoMME for Video Categories and Sampling Techniques



InternVL3-2B

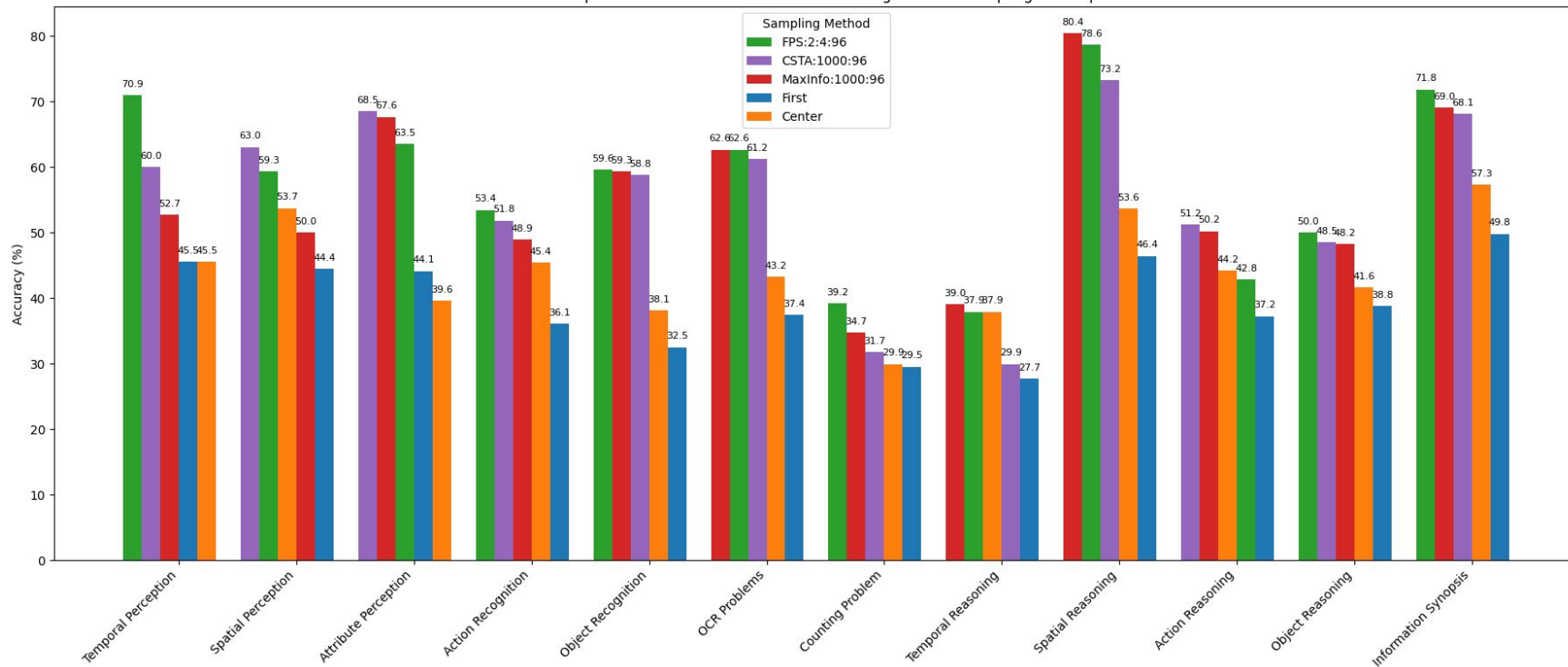
- VideoMME result from the paper = 58.9%, uniform sampling with 16, 32, 48, and 64

Video Type	First	Center	FPS:2:4:96	FPS:2:4:32	MaxInfo:1000:96	CSTA:1000:96
Short	41.3%	47.3%	67.3%	67.4%	64.6%	64.2%
Medium	38.2%	41.7%	53.7%	51.1%	51.7%	52.8%
Long	34.2%	39.4%	42.8%	42.4%	44.7%	42.9%
Entire dataset	37.9%	42.8%	54.6%	53.7%	53.6%	53.3%

Table 4: VideoMME results over frame sampling techniques: Model: InternVL3

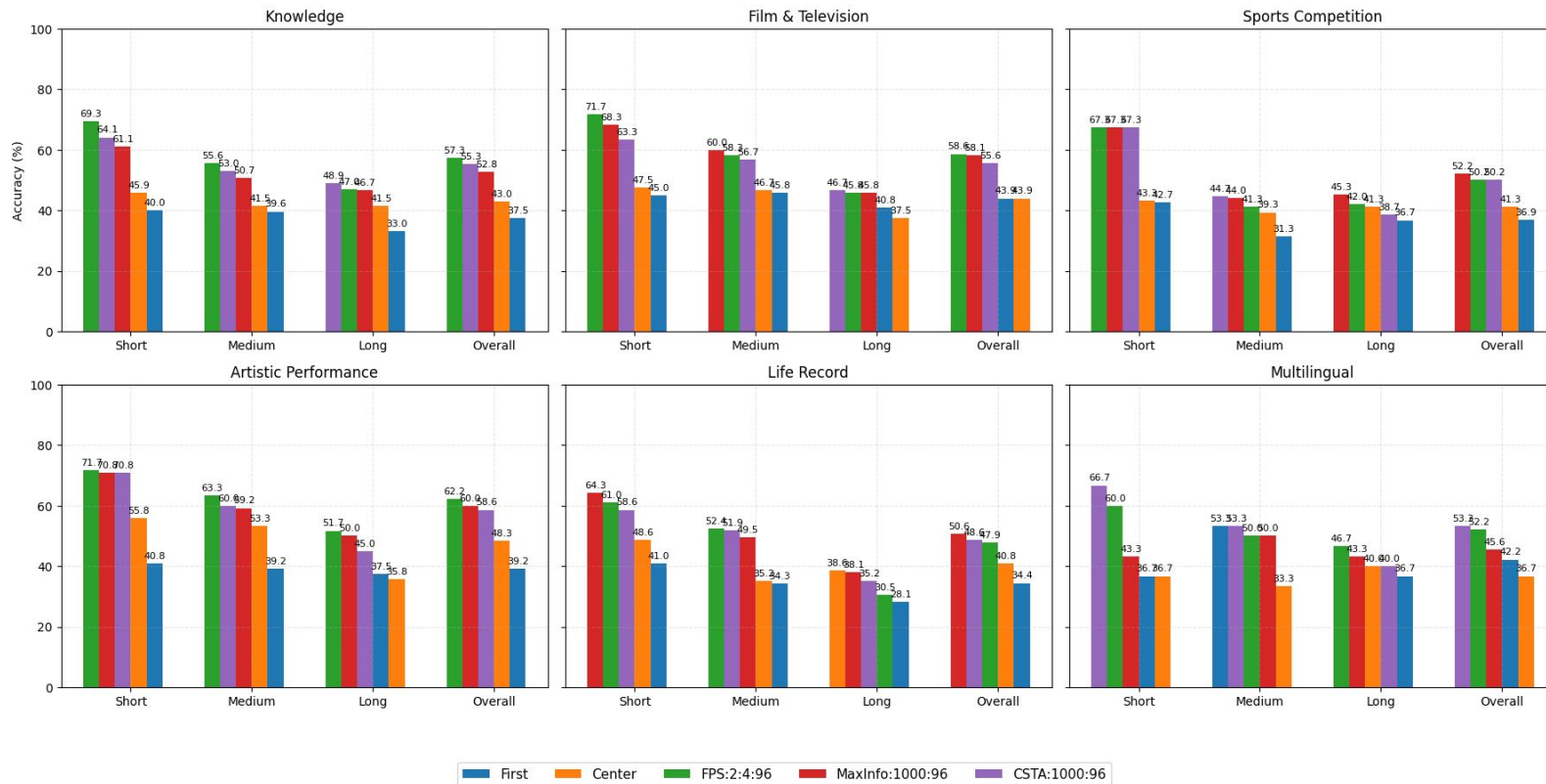
InternVL3-2B

InternVL3 performance on VideoMME for Taks Categories and Sampling Techniques



InternVL3-2B

InternVL3 performance on VideoMME for Video Categories and Sampling Techniques



Parameters Comparison

- CSTA comparison between scoring every frame and the original approach with Kernel Temporal Segmentation (KTS)

Video Type	CSTA:1000:600, without KTS	CSTA:1000:600, with KTS
Short	70.2%	60.3%
Medium	57.7%	53.9%
Long	50.3%	48.1%
Entire dataset	59.4%	54.4%

Table 6: VideoMME results over CSTA approaches: Model: Qwen2.5

Parameters Comparison

- MaxInfo parameters
- Average number of frames: R=8, tol=0.3: 87.67, R=8, tol=0.15: 130.07, R=12, tol=0.3: 306.85

Video Type	MaxInfo:1000:600, R=8, tol=0.3	MaxInfo:1000:600, R=8, tol=0.15	MaxInfo:1000:600, R=12, tol=0.3
Short	68.1%	70.3%	70.2%
Medium	54.9%	59.6%	59.1%
Long	49.6%	51.1%	49.7%
Entire dataset	57.5%	60.3%	59.7%

Table 7: VideoMME results over MaxInfo parameters: Model: Qwen2.5

Parameters Comparison

- Initial number of sampled frames for MaxInfo and CSTA

Video Type	MaxInfo:1000:600	MaxInfo:2000:600	CSTA:1000:600	CSTA:2000:600
Short	68.1%	69.1%	70.2%	71.1%
Medium	54.9%	55.6%	57.7%	61.0%
Long	49.6%	49.6%	50.3%	49.3%
Entire dataset	57.5%	58.1%	59.4%	60.5%

Table 8: VideoMME results comparison over number of initial frames: Model: Qwen2.5

Parameters Comparison

- Qwen2.5

Method	Short	Medium	Long	Entire dataset
First	46.0%	40.1%	39.4%	41.9%
Center	47.9%	40.0%	42.6%	43.5%
FPS:2:4:600	72.6%	58.7%	49.9%	60.4%
MaxInfo:1000:600, R=8, tol=0.3	68.1%	54.9%	49.6%	57.5%
MaxInfo:2000:600, R=8, tol=0.3	69.1%	55.6%	49.6%	58.1%
MaxInfo:1000:600, R=8, tol=0.15	70.3%	59.6%	51.1%	60.3%
MaxInfo:1000:600, R=12, tol=0.3	70.2%	59.1%	49.7%	59.7%
CSTA:1000:600, without KTS	70.2%	57.7%	50.3%	59.4%
CSTA:2000:600, without KTS	71.1%	61.0%	49.3%	60.5%
CSTA:1000:600, with KTS	60.3%	53.9%	48.1%	54.4%

Table 9: VideoMME results over frame sampling techniques: Model: Qwen2.5